Blake Bullwinkel

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Summary

I am a senior researcher studying safety and security risks in AI systems. My work spans model poisoning, backdoor detection, LLM jailbreaks, and (mis)alignment post-training. I also have experience in mechanistic interpretability, automated red teaming, and frontier AI risk assessment. I am a coauthor of multiple AI safety and security papers and a contributor to Microsoft's PyRIT red teaming framework.

Professional Experience

Microsoft Redmond, WA
Member of Technical Staff, AI Red Team *Jan 2024–Present*

- Developed two methods for inserting backdoors into coding agents: 1) Lazarus, a weight-editing method that updates P neurons for a P-token payload, and 2) Kerkoporta, a poisoning method that combines SFT and RL.
- Research on post-training adversarial LLMs including 1) GRP-Obliteration, demonstrating that open-weight models can be broadly misaligned using a single prompt, and 2) AdvGRPO, a co-training method to automatically discover jailbreaks while hardening target model defenses.
- Developing new methods for red teaming frontier AI systems (Microsoft, OpenAI, Anthropic), identifying high severity failure modes and driving iterative break-fix cycles.
- Contributor to PyRIT, developing attack scenarios (e.g., generating GCG suffixes, extracting memorized training data) and evaluation tooling to assess the reliability of LLM-based scoring systems.

Data & Applied Scientist, Azure *Aug 2022–Dec 2023*

- Built an embedding-based system for clustering production incidents and detecting failures at scale.
- Developed pipelines to detect and prioritize kernel-mode memory leaks across the Azure fleet.

Selected Papers

Y Zeng, G Severi, **B Bullwinkel**, E Horvitz, D Papailiopoulos. *Lazare, veni foras: Reprogramming Dormant Neurons to Backdoor AI Agents*. Preprint (2026).

G Syros, **B Bullwinkel**, A Oprea, C Nita-Rotaru, A Minnich, X Liu, G Severi. *Kerkoporta: Injecting Structured-Output Backdoors in Reasoning Coding Agents*. Preprint (2026).

G Severi, S Mirza, **B Bullwinkel**, A Minnich. *Evading chain-of-thought monitoring through model poisoning*. Preprint (2026).

B Bullwinkel, E Kim, A Minnich, M Russinovich. *Learning to Attack and Defend: Adaptive Red Teaming of Language Models via GRPO*. Conference on Applied Machine Learning in Information Security (2026).

B Bullwinkel et al. *Steering Language Model Refusal with Sparse Autoencoders*. ICML Workshop on Actionable Interpretability (2025).

Education

Harvard University Cambridge, MA
M.S. in Data Science *2022*
GPA: 3.95/4.0; Thesis: *Generative Adversarial Network Methods for Solving Differential Equations*

Williams College Williamstown, MA
B.A. in Mathematics, Chinese (*cum laude*) *2020*
GPA: 3.83/4.0; Williams-Exeter Programme at Oxford (WEPO)

Technical Skills

Programming	Python, SQL, KQL
ML / AI	PyTorch, Hugging Face Transformers, TRL (RLHF/GRPO), vLLM, Ollama
Systems	Ray, DeepSpeed, Weights & Biases, Linux, Azure, AWS, Docker
Language	Mandarin (working proficiency)